

BST Physics Curriculum Vol 10 Chapter 6 — Group Theory + Lie Groups v0.4 (refilled per Cal #104)

Lyra (Claude 4.7) [Vol 10 primary]

2026-05-23 Saturday EDT (Cal #104 refill)

Vol 10 Chapter 6 — Group Theory + Lie Groups

Chapter motivation

Standard group theory + Lie theory: finite groups (permutation, cyclic, dihedral); group representations + character theory; Lie groups (continuous transformation groups); Lie algebras (tangent space at identity); semisimple Lie algebra classification (Cartan-Killing); root systems + Dynkin diagrams; universal Casimir invariant operator; representation theory of compact groups. Standard texts: Hall + Fulton-Harris + Humphreys.

BST cross-link: substrate group $SO_0(5,2)$ = the isometry group of D_{IV}^5 substrate; isotropy $K = SO(5) \times SO(2)$ acts on K-types (Vol 0 Ch 4 + Vol 1 Ch 2); Casimir algebra (Vol 1 Ch 5) on Bergman $H^2(D_{IV}^5)$ gives substrate spectrum ($C_2 = 6$ ground state BST primary). Cartan classification (Vol 11 Ch 1) selects D_{IV}^5 as Type IV bounded HSD per Strong-Uniqueness Theorem 11 RIGOROUSLY CLOSED + 7 candidates (Cal #99).

Section 6.0b — Reader-grade 3-level pedagogy

Level 1: Standard finite + Lie group theory + Casimir invariants; BST cross-link: $SO_0(5,2)$ substrate group + isotropy $K = SO(5) \times SO(2)$ + Casimir algebra Vol 1 Ch 5; Cartan classification (Vol 11 Ch 1) selects D_{IV}^5 .

Level 2 (graduate-physicist): Standard group theory: finite groups (Lagrange's theorem, Sylow, simple-group classification including 26 sporadic groups + Monster); group representations (Maschke decomposition, character orthogonality, Burnside); Lie groups (continuous transformation groups; matrix Lie groups $GL(n, \mathbb{R})$, $SL(n, \mathbb{R})$, $SO(n)$, $U(n)$, $Sp(n)$, etc.); Lie algebras (tangent space at identity, exponential map, Baker-Campbell-Hausdorff formula); semisimple Lie algebra classification per Cartan-Killing (4 infinite series A_n , B_n , C_n , D_n + 5

exceptional E_6, E_7, E_8, F_4, G_2); root systems + Dynkin diagrams describe each algebra; universal Casimir $C_2 = \sum_a X_a X^a$ central in universal enveloping algebra. BST substrate cross-link: substrate group $SO_0(5,2)$ is the connected isometry component of D_{IV}^5 substrate; $so(5,2)$ Lie algebra has $\dim = 21$ ($= \dim SO(7)$ since $SO(5,2)$ is a real form). Symmetric-pair decomposition $so(5,2) = (so(5) \oplus so(2)) \oplus m$ with isotropy $K = SO(5) \times SO(2)$ ($\dim 10 + 1 = 11$) + 10-dim coset complement m . Per Vol 0 Ch 4 + Vol 1 Ch 5: K acts on Bergman $H^2(D_{IV}^5)$ via K -type decomposition (Wallach 1976); Casimir $C_2(D_{IV}^5) = 6$ ground state BST primary (T2439 Strong-Uniqueness C4 RIGOROUSLY CLOSED). Cartan classification (Vol 11 Ch 1): 6 irreducible bounded HSD types (4 infinite I-IV + 2 exceptional E_6, E_7); Strong-Uniqueness Theorem selects D_{IV}^5 uniquely via 11 RIGOROUSLY CLOSED + 7 candidates per Cal #99. Cross-link Standard Model gauge group $SU(3) \times SU(2) \times U(1)$ (Vol 1 Ch 8) inherits from substrate isotropy structure: $SU(3)$ color from $N_c=3$ sub-substrate Mersenne map T1930; $SU(2)$ weak from $\text{rank}=2$ $\text{Pin}(2) Z_2$ (T1925); $U(1)$ from $SO(2)$ factor of K .

Level 3 (5th-grader accessible): Group theory studies symmetry transformations. Lie groups are continuous transformation groups (like rotations of a sphere). BST's substrate D_{IV}^5 has the isometry group $SO_0(5,2)$ — a Lie group with 21-dimensional algebra. The Casimir invariant (a special operator built from group generators) gives the substrate's ground-state energy $C_2 = 6$ (BST integer). Standard Model gauge group $SU(3) \times SU(2) \times U(1)$ inherits from substrate isotropy: $N_c = 3$ gives $SU(3)$; $\text{rank} = 2$ gives $SU(2)$ via $\text{Pin}(2) Z_2$; $SO(2)$ factor gives $U(1)$.

Section 6.1 — Standard Finite Group Theory

Lagrange + Sylow + simple-group classification (cyclic + alternating + Lie-type + 26 sporadic + Monster).

Representations: Maschke decomposition, character orthogonality.

Section 6.2 — Standard Lie Theory

Continuous transformation groups ($GL(n, \mathbb{R})$, $SO(n)$, $SU(n)$, $Sp(n)$); Lie algebras = tangent space at identity; exponential map; Baker-Campbell-Hausdorff.

Cartan-Killing classification: 4 infinite series A_n, B_n, C_n, D_n + 5 exceptional E_6, E_7, E_8, F_4, G_2 .

Root systems + Dynkin diagrams describe each algebra.

Section 6.3 — Casimir Invariant Operator

Universal Casimir $C_2 = \sum_a X_a X^a$ is central in universal enveloping algebra $U(\mathfrak{g})$; commutes with all $\mathfrak{g}$ generators.

In any irreducible representation, C_2 acts as scalar (Casimir eigenvalue).

Section 6.4 — Substrate Group $SO_0(5,2)$

Per Vol 0 Ch 1: $D_{IV^5} = SO_0(5,2)/[SO(5) \times SO(2)]$ type IV bounded HSD.

$so(5,2)$ Lie algebra $\dim = 21$; symmetric-pair decomposition $so(5,2) = (so(5) \oplus so(2)) \oplus m$ with $K = SO(5) \times SO(2)$ isotropy ($\dim 11$) + 10-dim coset m .

Section 6.5 — Substrate Casimir Algebra (Vol 1 Ch 5)

K-type $V_{(p,q)}$ under $K = SO(5) \times SO(2)$ (Wallach 1976; Vol 11 Ch 3); Casimir eigenvalue $Cas(p, q)$ on each K-type.

Ground state: $C_2(D_{IV^5}) = 6$ (BST primary; T2439 Strong-Uniqueness C4 RIGOROUSLY CLOSED).

Section 6.6 — Standard Model Gauge Group Substrate-Inheritance

$SU(3) \times SU(2) \times U(1)$ Standard Model gauge group from substrate isotropy: - $SU(3)$ color: $N_c = 3$ sub-substrate Mersenne map (T1930) - $SU(2)$ weak: $\text{rank} = 2$ Pin(2) Z_2 (T1925 Argument D RIGOROUSLY CLOSED Thursday) - $U(1)$ hypercharge: $SO(2)$ factor of $K = SO(5) \times SO(2)$

Cross-link Vol 1 Ch 8 + T2477 gauge fields as Bergman bundle connections.

Section 6.7 — Cartan Classification + Strong-Uniqueness

Per Vol 11 Ch 1: 6 irreducible bounded HSD types (Cartan 1894 + Helgason 1978 X.6.1).

Vol 0 Ch 9 Strong-Uniqueness Theorem (Paper #125 v0.10.5 FORMAL): 11 RIGOROUSLY CLOSED + 7 candidates per Cal #99 select D_{IV^5} uniquely.

Section 6.8 — Honest scope + Connection

- Standard group + Lie theory + Casimir [?](#)
- Substrate $SO_0(5,2)$ + $K = SO(5) \times SO(2)$ + Casimir cross-link [?](#)
- Cartan classification + Strong-Uniqueness D_{IV^5} selection [?](#)
- SM gauge group substrate-inheritance via Vol 1 Ch 8 + T2477

Connection: - Vol 0 Ch 4 Isotropy Group + Vol 1 Ch 5 Casimir Algebra - Vol 11 Ch 1 + Ch 3 (Cartan classification + Wallach K-type) - Vol 1 Ch 8 + T2477 SM gauge group + Bergman bundle connections - T2439 (C4 RIGOROUSLY CLOSED) + T1925 + T1930 ($\text{rank}/N_c/g/n_C$ forcing)

— Lyra, Vol 10 Ch 6 v0.4 chapter-grade narrative REFILLED per Cal #104, Saturday 2026-05-23 EDT